



E-Commerce Customer Churn Prevention

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Data Overview

- We found a churn dataset on Hugging Face. 36,992 customer records.
 - 24 total columns

Split (1)
train · 37k rows

Search this dataset

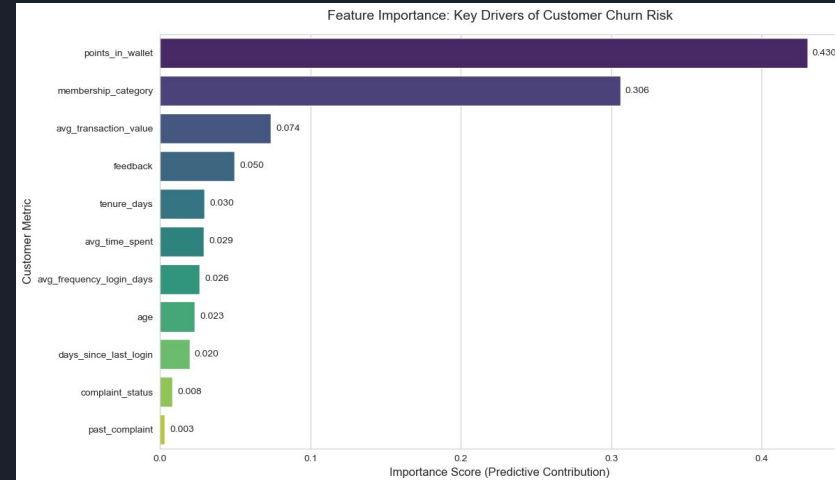
special_discount	offer_application_preference	past_complaint	complaint_status	feedback	churn_risk_score
string · classes	string · classes	string · classes	string · classes	string · classes	int64
45%	Yes 55.3%	Yes 49.7%	Solved in 12.4%	Poor Website 17%	1 54.1%
	Yes	No	Not Applicable	Products always in Stock	0
	No	Yes	Solved	Quality Customer Care	0
	Yes	Yes	Solved in Follow-up	Poor Website	1
	Yes	Yes	Unsolved	Poor Website	1

- + Target Variable: **churn_risk_score** (Binary: 0 for Stay, 1 for Churn).

Mean Churn Risk for Dataset: **0.541**

Main Feature Categories:

- Financial: Points in wallet, average transaction value.
- Engagement: Days since last login, time spent on platform, tenure.
- Sentiment/Emotions: Customer feedback labels and complaint status.



Methodology

01

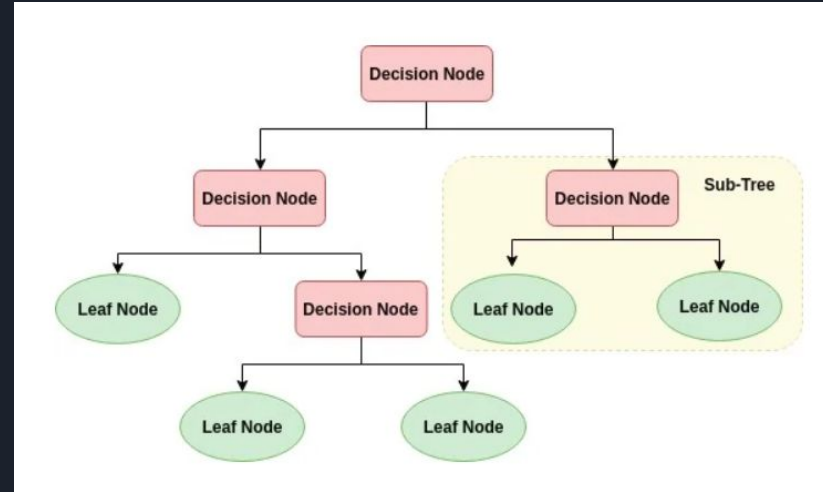
Preprocessing and Data Exploration:
Valid inputs for our ML models and see important features for churn in this specific dataset.



Methodology

02 **ML Models:** HistGradientBoosting, Random Forest, and a Gradient Boosting Classifier. Optimize for Recall

03 **Evaluation and Strategy:** View Precision, Recall, and F1-score of models. Highlight high churn probabilities and segment them.



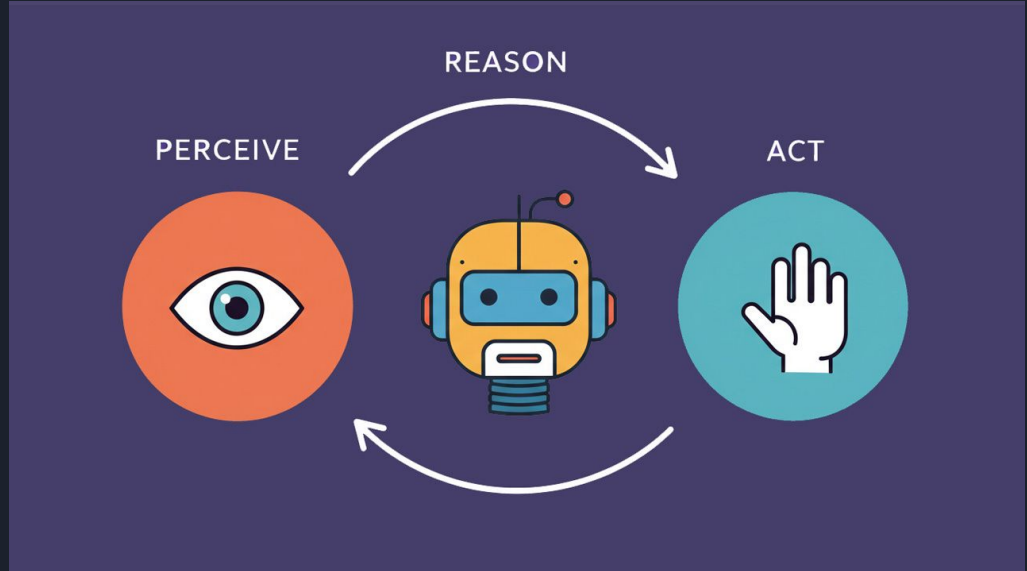
Segment	Count
Standard Retarget	16,793
Loyalty Rewards	13,885
Upsell Opportunity	3,425
VIP Recovery	2,889

Methodology

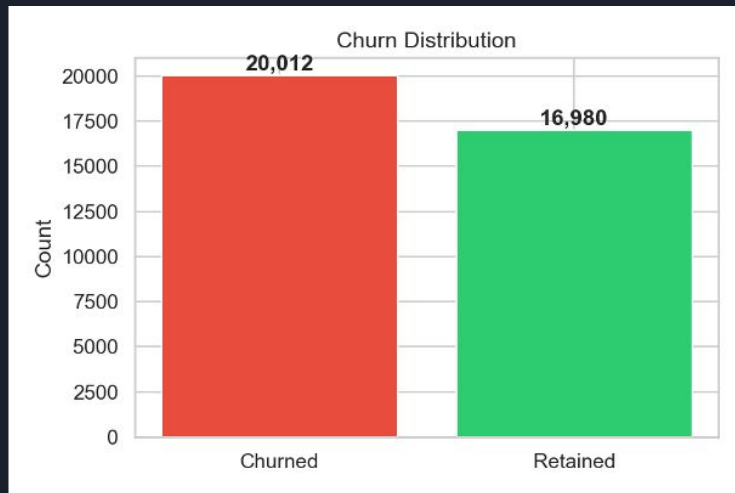
04

AI Agents:

- A Strategy agent
- A Critic agent
- An Executor Agent

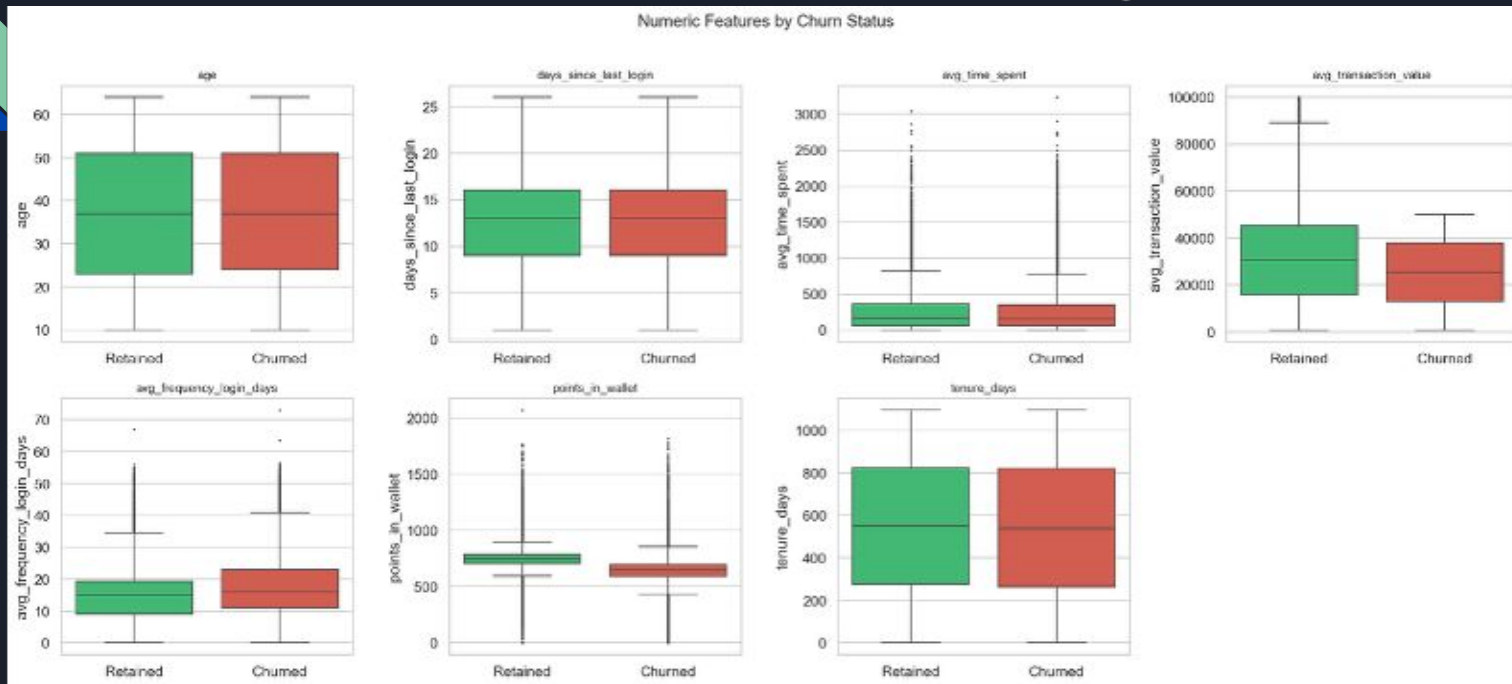


EDA: What Drives Churn?



- 54.1% overall churn rate (20,012 churned vs 16,980 retained) — balanced for ML
- Membership category is the #1 predictor: categories 0 & 2 → ~97% churn; 3 & 4 → 0% churn
- Upgrading customers to higher membership tiers is the highest-ROI retention lever

EDA: Feature Analysis

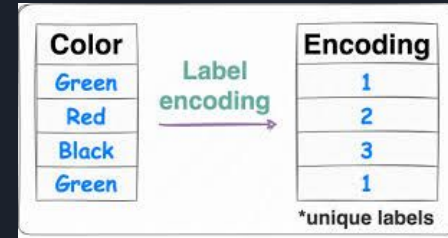


- Retained customers have higher transaction values and more wallet points
- Age, tenure, days since login show virtually no difference
- support_risk and past_complaint are perfectly correlated
- Monetary engagement separates churners from retained

Preprocessing Step

- **Data Validation:** Cleaned up the features by identifying and fixing invalid placeholders
- **Feature Engineering:**
 - Make tenure_days
 - Encoded categorical features

points_in_wallet	used_special_discount	offer_application_preference	past_complaint	complaint_status	feedback
781.750000	Yes		Yes	No	Products always in Stock
NaN	Yes		No	Yes	Quality Customer Care



- **Noise Reduction:**
 - Dropped non-predictive identifiers (referral_id)
 - Dropped unneeded features

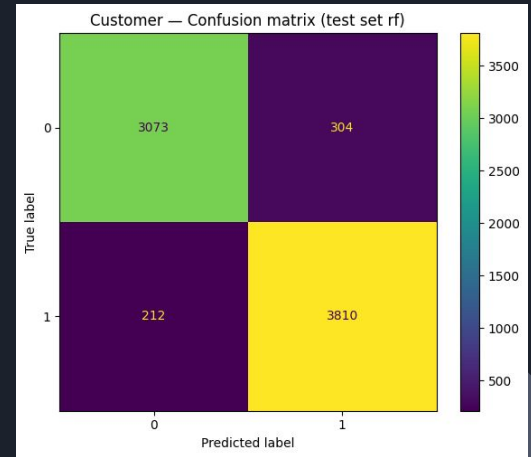
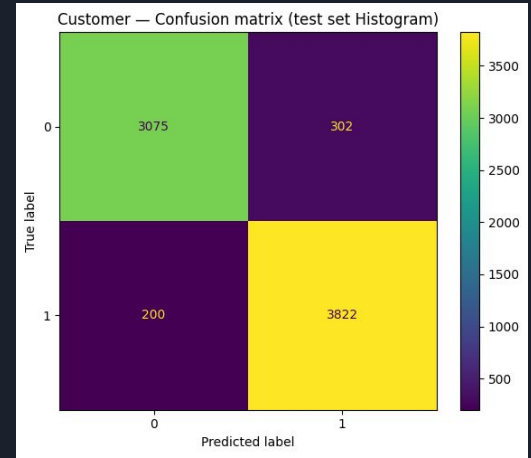
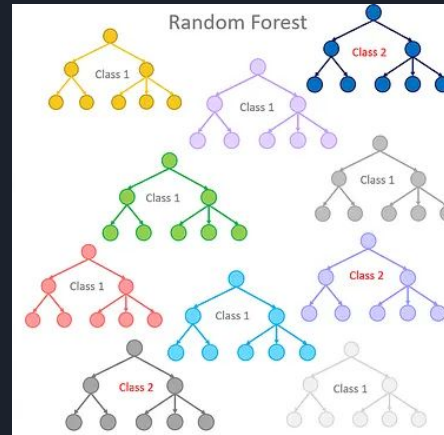
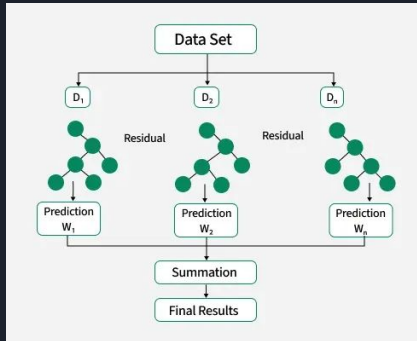


age	gender	referral_id	region_category
18	F	xxxxxxxx	Village
32	F	CID21329	City
44	F	CID12313	Town
37	M	CID3793	City
31	F	xxxxxxxx	City

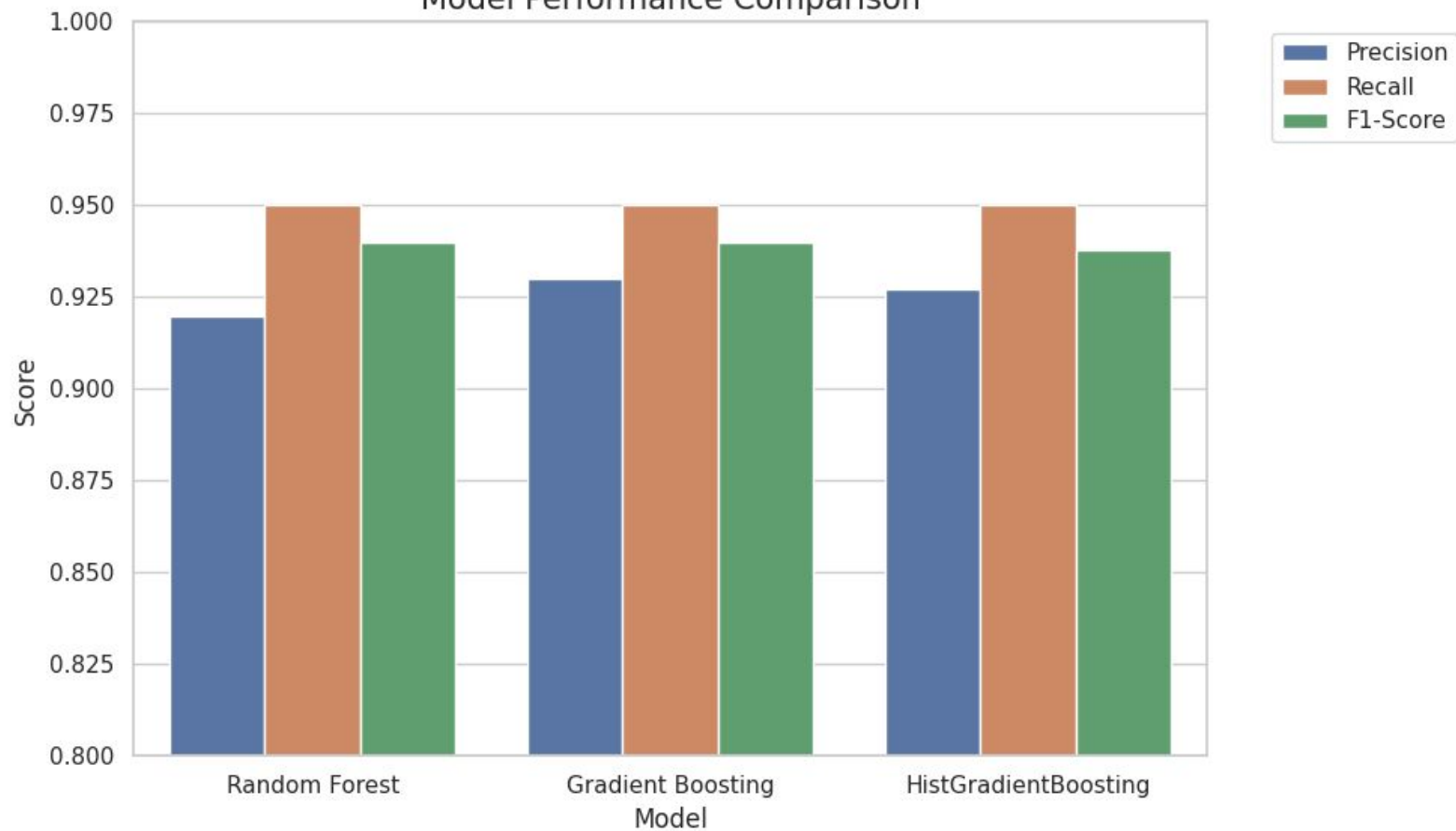
Training our ML Models

Churn patterns are complex and nonlinear. These are good at looking at customer behavioral data

- HistGradientBoosting
- Random Forest
- Gradient Boosting
- XGBoost



Model Performance Comparison





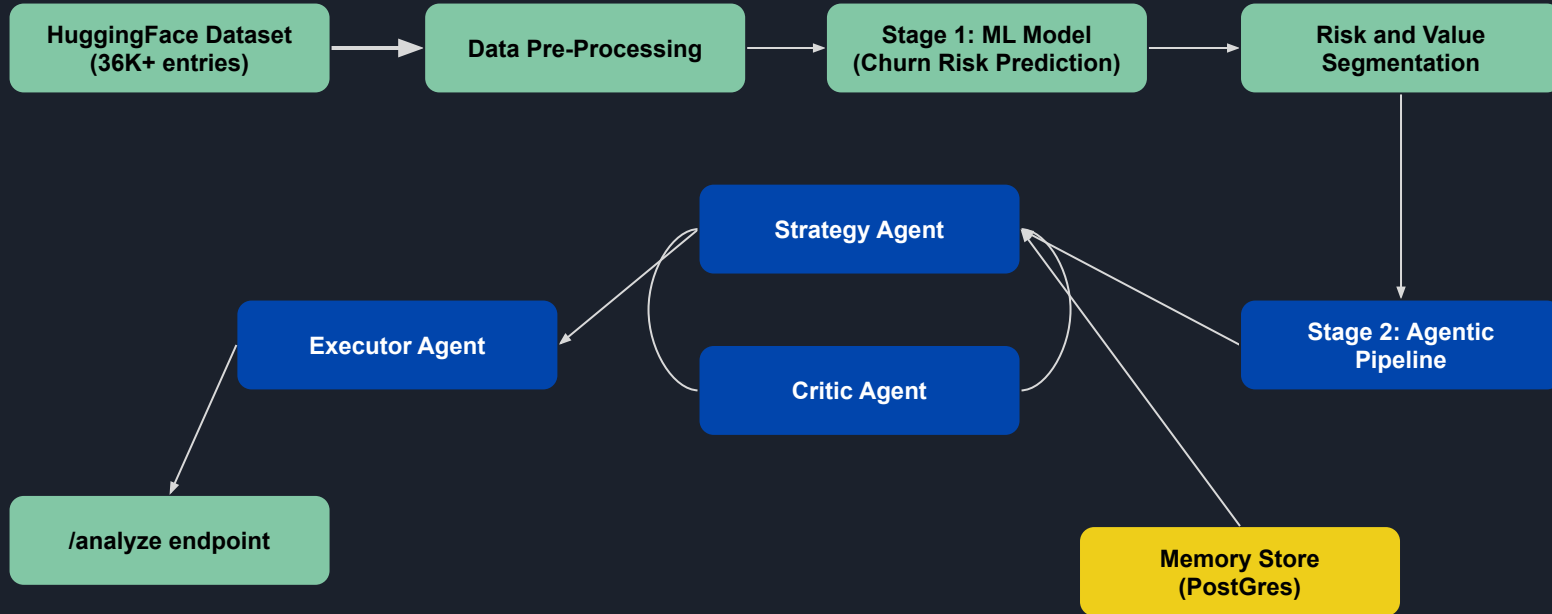
Our Segmentation and Prevention Strategy

- Take our churn probabilities from best model and extract the high risk customers (>70% chance)
- Segment the customers by “priority”

	security_no	churn_probability	segment
0	NK8DB3U	0.990239	VIP-At-Risk
1	YN8ZX7A	0.989415	Standard-At-Risk
2	W6D4LHA	0.988971	Standard-At-Risk
3	4XMZGUM	0.988696	Standard-At-Risk
4	2W6QNJE	0.988523	Standard-At-Risk
5	YBFSWXJ	0.988389	VIP-At-Risk
6	YG4KTEH	0.988195	Standard-At-Risk
7	JO6QG54	0.988184	VIP-At-Risk
8	W7EEJUC	0.988147	Standard-At-Risk
9	3UVF5Q2	0.988128	Standard-At-Risk

	feedback_text	churn_risk_score
2	Poor Customer Service	0.649921
1	Too many Ads	0.638516
0	Poor Product Quality	0.634976
3	Poor Website	0.631797
7	User Friendly Portal	0.627011
4	No reason specified	0.000000
5	Good Product Quality	0.000000
6	Reasonable Price	0.000000
8	Products always in Stock	0.000000

Agentic Churn Prevention Pipeline





Agentic Pipeline: The Debate Mechanism

Round 1: Strategist Proposes

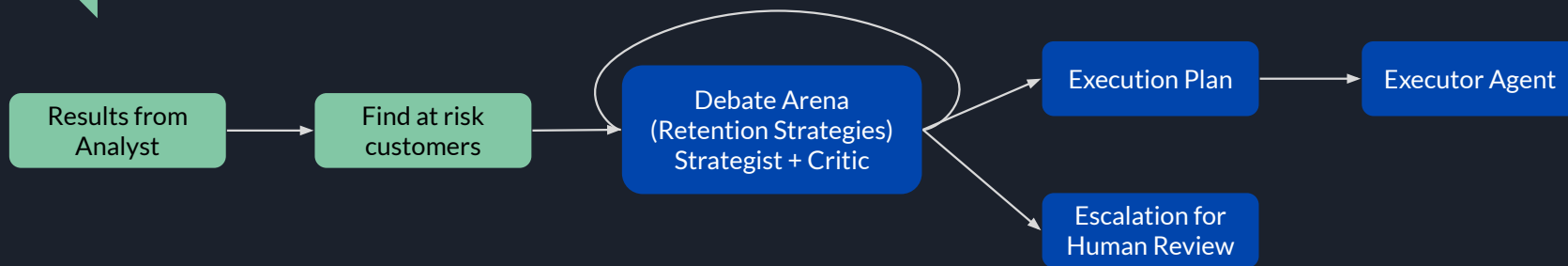
- Generic email campaigns & loyalty discounts
- Personalized points showcasing
- FAQ guides for rewards programs

Round 2: Strategist Revises

- Validation-first: survey to check model calibration
- A/B test: 10% discount vs feature highlight
- Personalized points redemption with control group

What is the intuition?

This is the logical flow of our Churn prevention agentic pipeline



- Analyst report (Results from Analyst)
- Output of Debate Arena?
- How the debate is stored?



API Structure - **/analyze**

API Request Structure

```
postman request POST 'http://localhost:8000/api/v1/analyze' 'file=@"/Users/raghu/coding/ECE143/ece143-final-project/churn-prevention-agents/test_data.csv"'
```

API Response Structure

```
{  
  "job_id": "1fad9e06-35c1-4739-81e8-d6bf62edc525",  
  "status": "processing",  
  "message": "Pipeline started"  
}
```



API Response Structure - **/results**

API Request Structure

```
postman request 'http://localhost:8000/  
api/v1/results/  
1fad9e06-35c1-4739-81e8-d6bf62edc525'
```

API Response Structure (Debating...)

```
{  
  "job_id": "1fad9e06-35c1-4739-81e8-d6bf62edc525",  
  "status": "debating",  
  "analyst_report": null,  
  "debate_rounds": 0,  
  "approved_plan": null,  
  "execution_package": null,  
  "debate_log": null,  
  "created_at": "2026-03-11T04:21:07.020731",  
  "completed_at": null  
}
```

API Response Structure - /results

```
{
  "job_id": "1fad9e06-35c1-4739-81e8-d6bf62edc525",
  "status": "escalated",
  "analyst_report": {
    "segments": [ ... ],
    "total_at_risk": 6,
    "global_top_drivers": [ ... ]
  },
  "debate_rounds": 5,
  "approved_plan": null,
  "execution_package": null,
  "debate_log": { ... },
  "created_at": "2026-03-11T04:21:07.020731",
  "completed_at": "2026-03-11T04:25:48.249256"
}
```

API response body structure

```
"debate_log": {
  "proposals": [
    {
      "proposals": [
        {
          "segment_id": "driver_TechSupport",
          "action_type": "proactive_call",
          "offer_details": "Dedicated support specialist call within 24h to resolve pending technical issues and offer proactive assistance.",
          "channel": "phone",
          "timing": "within_24h",
          "estimated_cost_per_customer": 15.0,
          "expected_retention_lift": 0.12,
          "estimated_roi": 3.48,
          "rationale": "Addresses the root cause directly with personalized support to resolve issues, rebuild trust, and prevent churn due to technical frustrations."
        },
        {
          "segment_id": "driver_MonthlyCharges",
          "action_type": "discount",
          "offer_details": "15% discount on the next 3 months' bill, communicated via SMS with a direct link to claim.",
          "channel": "sms",
          "timing": "immediate",
          "estimated_cost_per_customer": 25.0,
          "expected_retention_lift": 0.15,
          "estimated_roi": 1.47,
          "rationale": "Directly addresses price sensitivity by offering a tangible financial incentive to reduce perceived cost and provide immediate relief."
        }
      ]
    }
  ]
}
```

Debate Logs that show agent debate



API Response Structure - **/results**

```
"critiques": [  
  {  
    "segment_evaluations": [  
      {  
        "segment_id": "driver_TechSupport",  
        "objections": [  
          {  
            "severity": "MODERATE",  
            "issue": "ROI Realism: The expected retention lift of 12% is not substantiated by any historical data. While the intervention is strong, the estimate is theoretical and carries a risk of being overestimated.",  
            "suggestion": "Run this as a pilot with a control group to establish a baseline lift for this action before a full-scale rollout. This will validate the ROI projection."  
          }  
        ],  
        "rating": 8  
      },  
      {  
        "segment_id": "driver_MonthlyCharges",  
        "objections": [  
          {  
            "severity": "CRITICAL",  
            "issue": "ROI Realism: An expected retention lift of 15% for a standard discount offer is highly optimistic without historical data to support it. The projected ROI of 1.47 is entirely dependent on this unverified assumption.",  
            "suggestion": "Launch as an A/B test against a control group and a lower-cost alternative (e.g., a 10% discount or a one-time credit) to measure the actual lift and cost-effectiveness."  
          }  
        ],  
        "rating": 5  
      }  
    ]  
  },  
  {  
    "segment_evaluations": [  
      {  
        "segment_id": "driver_TechSupport",  
        "objections": [  
          {  
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          }  
        ],  
        "rating": 5  
      }  
    ]  
  }  
]
```

Agent Critiques in the Debate Logs



Key Takeaways

1

Membership tier is the #1 churn lever

46% of model importance — upgrade programs have highest ROI

2

XGBoost achieves 97.6% AUC

Confirms EDA: membership + wallet points = 67% of decisions

3

Feedback is a near-perfect early warning

Types 4-6, 8 → 0% churn; types 0-3, 7 → 63% churn

4

Agentic pipeline bridges prediction → action

Multi-agent debate ensures strategy quality (3/10 → 8/10)

5

End-to-end: data → model → agents → campaigns

Fully automated, API-driven, Docker-deployed



Thank you!